Dialogue Act Analysis of Facilitator-Children Multilingual Communication

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01 Introduction

◆SDGs「Quality Education」

- To develop global citizenship, it is important to appreciate cultural diversity.

◆Children experience intercultural collaboration using machine translation

- To understand the others
- To make themselves understood

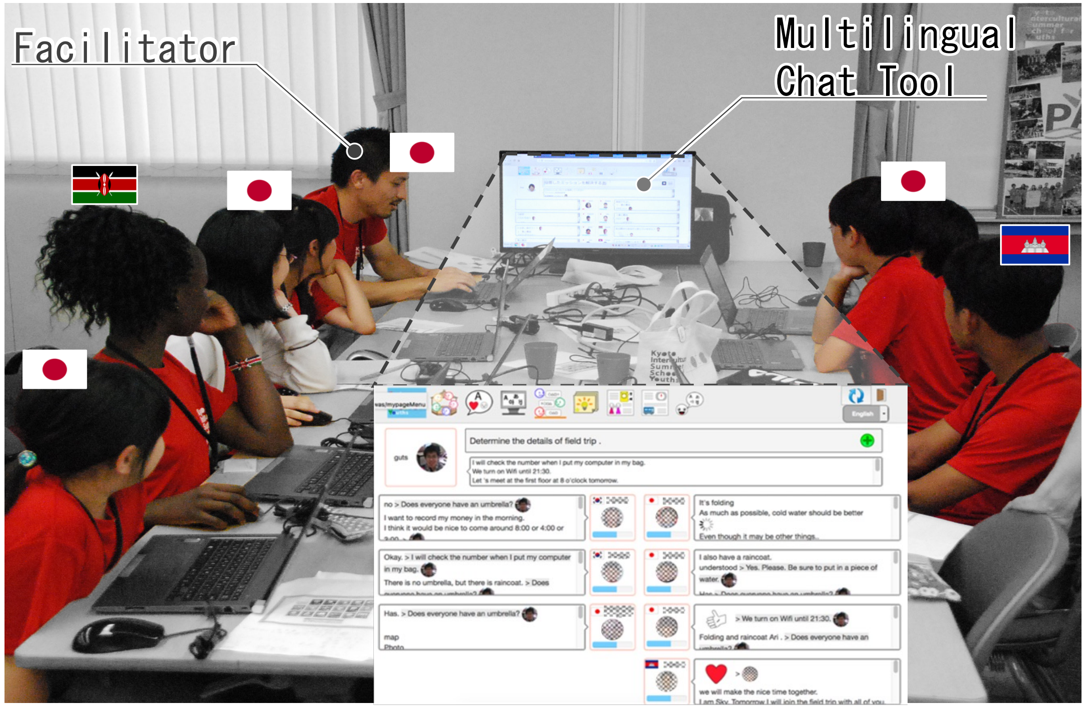
4 QUALITY EDUCATION

SUSTAINABLE DEVELOPMENT GOALS

01 Introduction

◆Children's Intercultural Collaboration

- KISSY(Kyoto International Summer School for Youth)

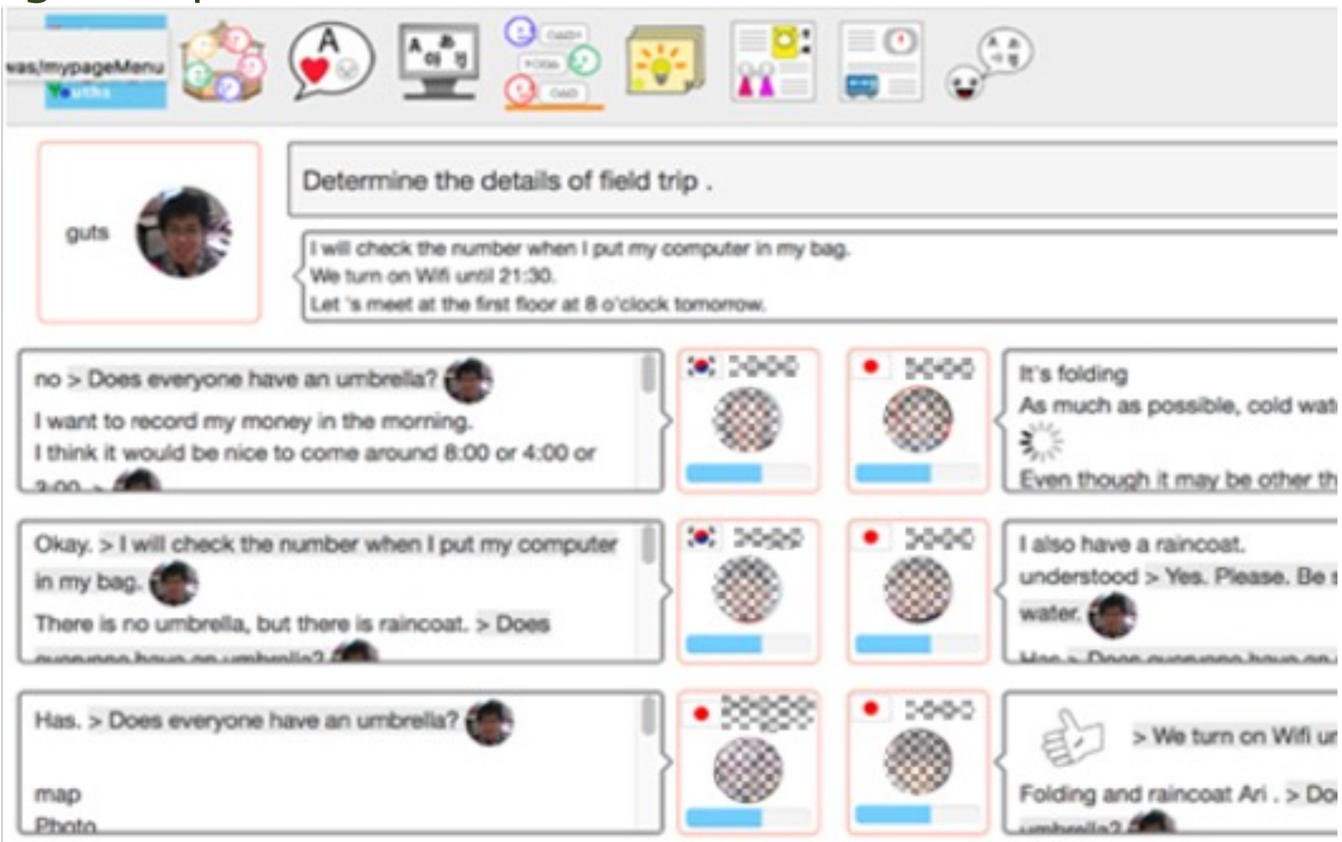


Source: PANGAEA

01 Introduction

◆Chat tool used in KISSY

- English speaker screen



01 Introduction

◆Chat tool used in KISSY

- Japanese speaker screen

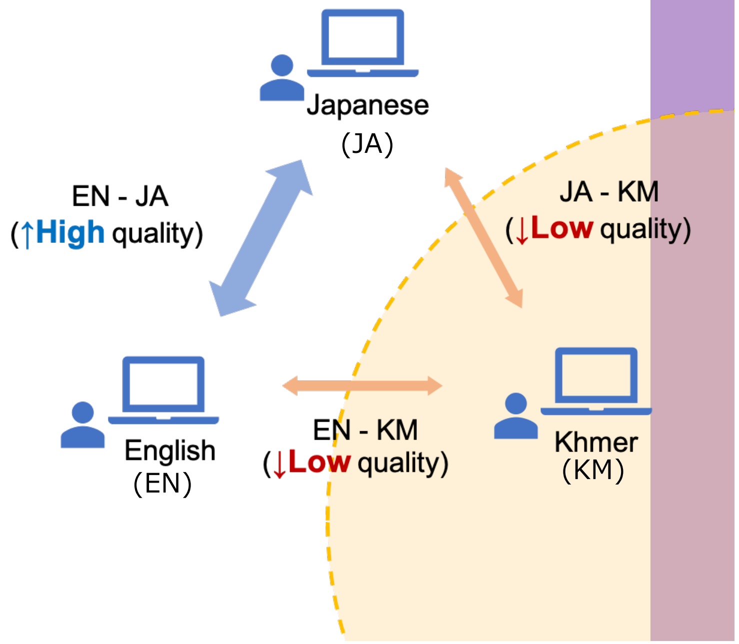
02 Purpose

◆ Children speaking low-resource languages tend to speak less[1]

- **low-resource languages**: Languages with fewer language resources available for machine translation and lower translation accuracy.

◆ Focus on the facilitator’s dialogue acts that children receive.

◆ Clarify the facilitation that children of low-resource languages are likely to utter and the effects of machine translation on the dialogue acts.



[1] Mondheera Pituxcoosuvarn, Toru Ishida, Naomi Yamashita, Toshiyuki Takasaki, Yumiko Mori (2018): Machine Translation Usage in a Children’s Workshop: Proceesings of The Eighth International Conference on Collaboration Technologies (CollabTech’ 18): pp.59-73

03 Hypotheses

H1

Does each dialogue act affect the response for each language user?

H2

Does the number of responses to each dialogue act differ across languages?

H3

How does machine translation affect the facilitator’s dialogue act?

04 Utterance Annotation

◆Log data was tagged using ISO 24617-2 and quantified

Dialogue Act	Tag Definition
Information-Providing (IP)	The utterance of a dialogue act in which the speaker aims to inform the listener of certain information.
Information-Seeking (IS)	The utterance in which the speaker questions the listener to get the information that he is looking for.
Directive	The utterance in which the speaker puts pressure on the addressee to perform a certain action.
Commissive	The utterance in which the speaker commits oneself to perform a certain action.
Dimension-Specific (DS)	The utterance in which the speaker performs social obligations, for example, greeting, expressing her/his gratitude, and apologizing.

04 Utterance Annotation

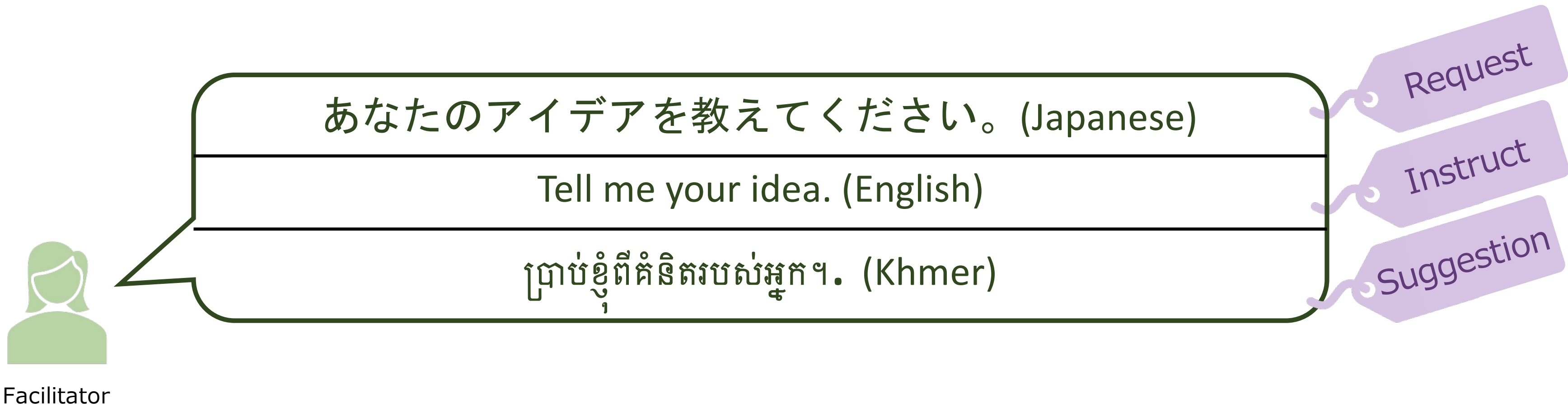
◆Also used the tag set of subdivided “Directive"

Dialogue Act	Tag Definition
Request	The directive in which the speaker performed to make the addressee conditional on the addressee’s consent to perform the action.
Instruct	The directive in which the speaker performed to make the addressee regardless of consent to perform the action.
Suggestion	The utterance in which the speaker performed to make the addressee consider the performance of a certain action.
Address-Offer	The utterance which the speaker performed indicates the possibility that listeners perform the action that the speaker offered to perform.

04 Utterance Annotation

◆Tagging facilitator utterances in KISSY log data

- Japanese, English, and Khmer (low-resource language)
- 3-4 annotators for each language



05 Analysis of Hypothesis 1

The number of responses and non-responses to each dialogue act by language used.

Dialogue Act	Japanese		English		Khmer	
	Response	Non-Response	Response	Non-Response	Response	Non-Response
Information-seeking	16	54	13	38	8	12
Request	50	130	2	9	0	1
Instruct	1	24	4	16	0	1
Suggestion	3	52	2	12	0	7

◆A significance test was conducted to clarify the effect of each dialogue act on the number of responses.

◆The data of Khmer “Request” and “Instruct” were not included in the analysis.

05 Result of Hypothesis 1

Significance level: 0.05

Language	p-value[1]	Residual Analysis[2]
Japanese	P < .001 **	Request(+), Instruct(-), Suggestion(-)
English	P = .905 ns	
Khmer	P = .068 ns	

1. ns:p ≧ .050, *:p < .050, **:p < .010
2. (+) Significant in Number of Responses, (-) Significant in Number of Non-Responses

Discussion

◆ The dialogue acts that affect response are differ depending on the language used.

06 Analysis of Hypothesis 2

The number of responses and non-responses to each language

	Information-Seeking		Request		Instruct		Suggestion	
Language	response	non-response	response	non-response	response	non-response	response	non-response
Japanese	16	54	50	130	1	24	3	52
English	13	38	2	9	4	16	2	12
Khmer	8	12	0	1	0	1	0	7

- ◆A significance test was conducted to clarify the effect of each language used on the number of responses.
- ◆The data of Khmer “Request” and “Instruct” were not included in the analysis.

06 Result of Hypothesis 2

Significance level: 0.05

Utterance Type	P-value
Information-Seeking	P = .303
Request	P = .730
Instruct	P = .155
Suggestion	P = .423

Discussion

- ◆ There were no significant differences in all dialogue act.
- ◆ Japanese users did not respond to them significantly more or less than other language users.
- ◆ Depending on the language used, there may be differences in the number of times each dialogue act is received.

07 Analysis of Hypothesis 3

The number of receives each language used by dialogue act

Language used	IP[1]	IS[2]	Request	Instruct	Suggestion	Commissive	DS[3]
Japanese	32	14	36	5	11	2	19
English	35	37	7	14	10	12	4
Khmer	46	34	4	2	9	3	15

1: Information-providing, 2: Information-seeking, 3: Dimension-specific

- ◆Focused on the number of times each dialogue act is received.
- ◆A significant difference test was conducted to clarify the bias in the dialogue acts received by each language user.

07 Result of Hypothesis 3

Language used	IP[1]	IS[2]	Request	Instruct	Suggestion	Commissive	DS[3]
Japanese	32	14(-)	36(+)	5	11	2	19(+)
English	35	37(+)	7(-)	14(+)	10	12(+)	4(-)
Khmer	46(+)	34	4(-)	2	9	3	15

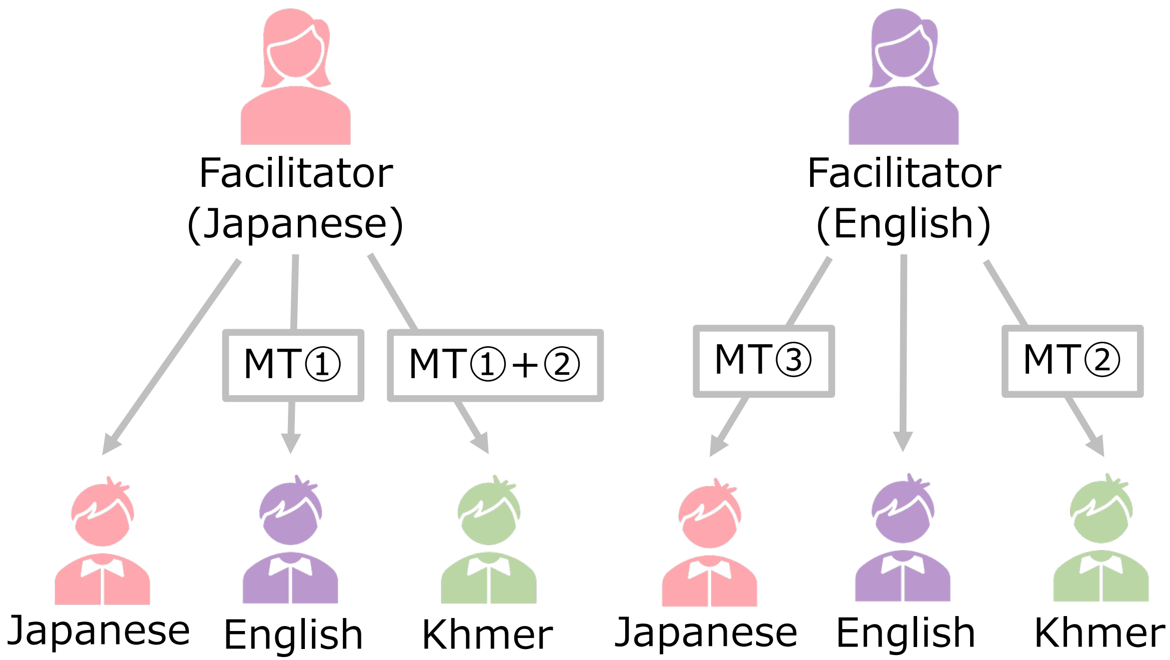
1: Information-providing, 2: Information-seeking, 3: Dimension-specific

Discussion

- ◆There were the significant differences in some dialogue acts.
- ◆This suggests that the dialogue act was changed by machine translation.

07 Analysis of Hypothesis 3

We need extract the language pair to analyze changes in dialogue act before and after translation.



Language pair

① Japanese to English

② English to Khmer

③ English to Japanese

07 Analysis of Hypothesis 3

The number of changes in dialogue acts before and after translation by each language pair.

<Part of the tabulation table(Japanese to English)>

		After Translation(English)						
		IP[1]	IS[2]	Request	Instruct	Suggestion	Commissive	DS[3]
Before Translation (Japanese)	IP[1]	16	1	0	0	1	0	0
	IS[2]	0	10	1	0	1	0	0
	Request	0	15	5	4	0	0	0
	Instruct	0	0	0	3	0	0	0
	Suggestion	0	0	0	3	3	1	0
	Commissive	0	0	0	0	0	1	0
	DS[3]	1	1	1	0	0	0	10

1: Information-providing, 2: Information-seeking, 3: Dimension-specific

07 Analysis of Hypothesis 3

The number of changes in dialogue acts before and after translation was extracted for each dialogue act.

<An example of the table for McNemar test (Japanese to English / Information-seeking)>

		After Translation(English)	
		Information-seeking	**Others**
Before Translation (Japenese)	**Information-seeking**	10	2
	Others	17	49

◆A McNemar test was conducted to clarify changeable dialogue acts in each language pair.

07 Result of Hypothesis 3

Language Pair	Result of McNemar Test (Type of Change[1])	
	Change to other dialogue acts	Change from other dialogue acts
Japanese to English	Request	Information-Seeking, Instruct
English to Khmer	Information-Seeking	Request
English to Japanese	Instruct	Information-Providing

Discussion

- Specific dialogue acts were found to be changed before and after translation.
- **Between Japanese and English**: there were changes between dialogue acts that need a response.
- **For English to Khmer**: the dialogue act changed from one that need a response to one that did not need a response.

→ This could be a factor that inhibits response in low-resource language user (Khmer)

08 Discussion

◆Examples of Changes in Dialogue Acts

- "Request"(Japanese) → "Information-seeking"(English)

移民を学校に受け入れるための解決策についてアイデアはありますか?

Do you have any ideas on solutions to accommodate immigrants in our schools?
(**Please tell me know if you have**.)

Facilitator
(Japanese)

Translation to English

Do you have an idea for a solution to accept immigrants in school?

- "Instruct"(English) → "Information-providing"(Khmer)

About how the story ends. I want the original blue team idea!

Facilitator
(English)

Translation to Khmer

អំពីរបៀបដែលរឿងបញ្ចប់។ ខ្ញុំចង់បានគំនិតក្រុមខៀវដើម!
(I want the blue team's original idea how to end the story!)

08 Design Implications

Machine Translation

◆ The machine translation may not always correctly convey the sender's intent

◆ It is necessary to take into consideration **whether the sender's intention is correctly received** to expand machine translation

Facilitation Method

◆ The utterances that changed the dialogue act were indirect expressions used by the facilitator.

◆ Facilitator **should avoid using indirect directive** in multilingual communication using machine translation.

- Ex: “Please tell me your idea.” is better than “Do you have any ideas?”

Thank you for listening !

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